Roof Layout Designer

18 modules / 7.38 kW

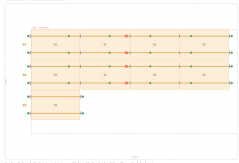
ROOF TYPE: Gable | Hip | Flat | Shed | Ground | + ADD FACE

FASTENING: HALO | MODULE ORIENTATION: Mixed | MODULE GAP (IN): 0.7 | ORIENT: Flush/Parallel / 40° open

South Face

AZ 180 | TILT 6° (Flat) | MPS Landscape | WIDTH 30 | DEPTH 18 | 9 mods / 3.7 kW

Auto-Fill | Reset Positions | ← 100% + FIT



Module: 71.8" x 1.2" (landscape) Array: 12.0' x 18.0' Scale: 10ft = 20px 9 of 9 placed

Key: Key | Valid | Bad Zone | Set Zone
STRINGS: S1

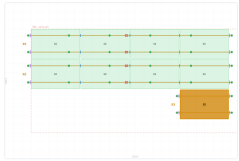
RACKING — 9 MODULES, 2 ROWS

ZONE	MODS	ANCHS	DEVICE	DEVICES	FOOT	RAIL	END
R1	4	273.5"	4 pcs - 2 rows 2-2 pcs spliced (110" + 163.5" per row, from 110" + 110" stock)	2	12	4	4
R2	4	273.5"	4 pcs - 2 rows 2-2 pcs spliced (110" + 163.5" per row, from 110" + 110" stock)	2	12	4	4
R3	1	47.5"	2 pcs - 2 rows 1-1 pc (47.5" cut from 110" stock)	-	4	2	2

West Face

AZ 180 | TILT 6° (Flat) | MPS Landscape | WIDTH 30 | DEPTH 18 | 9 mods / 3.7 kW

Auto-Fill | Reset Positions | ← 100% + FIT



Module: 71.8" x 1.2" (landscape) Array: 30.0' x 18.0' Scale: 10ft = 20px 9 of 9 placed

Key: Key | Valid | Bad Zone | Set Zone
STRINGS: S2

RACKING — 9 MODULES, 2 ROWS

ZONE	MODS	ANCHS	DEVICE	DEVICES	FOOT	RAIL	END
R1	4	273.5"	4 pcs - 2 rows 2-2 pcs spliced (110" + 163.5" per row, from 110" + 110" stock)	2	12	4	4
R2	4	273.5"	4 pcs - 2 rows 2-2 pcs spliced (110" + 163.5" per row, from 110" + 110" stock)	2	12	4	4
R3	1	47.5"	2 pcs - 2 rows 1-1 pc (47.5" cut from 110" stock)	-	4	2	2

LAYOUT SUMMARY: 2 faces | 18 modules | 7.38 kW | Sizing: 5/1.+50.0% (+50%)

Rail Cut List

#	STOCK NO.	FROM/OF	LEN	WASTE
1	96" 1M-1M5-1A6-B1	146.5" South Face R1	146.5"	24.5"
2	96" 1M-1M5-1A6-B1	146.5" South Face R1	146.5"	24.5"
3	96" 1M-1M5-1A6-B1	146.5" South Face R2	146.5"	24.5"
4	96" 1M-1M5-1A6-B1	146.5" South Face R2	146.5"	24.5"
5	96" 1M-1M5-1A6-B1	146.5" West Face R1	146.5"	24.5"
6	96" 1M-1M5-1A6-B1	146.5" West Face R1	146.5"	24.5"
7	96" 1M-1M5-1A6-B1	146.5" West Face R2	146.5"	24.5"
8	96" 1M-1M5-1A6-B1	146.5" West Face R2	146.5"	24.5"
9	132" 1M-1M5-1C3G1	132" South Face R1	132.0"	0.0"
10	132" 1M-1M5-1C3G1	132" South Face R1	132.0"	0.0"
11	132" 1M-1M5-1C3G1	132" South Face R2	132.0"	0.0"
12	132" 1M-1M5-1C3G1	132" South Face R2	132.0"	0.0"
13	132" 1M-1M5-1C3G1	132" West Face R1	132.0"	0.0"
14	132" 1M-1M5-1C3G1	132" West Face R1	132.0"	0.0"
15	132" 1M-1M5-1C3G1	132" West Face R2	132.0"	0.0"
16	132" 1M-1M5-1C3G1	132" West Face R2	132.0"	0.0"
17	204" 1M-1M5-2H4G1	47.5" South Face R6 47.5" South Face R6 47.5" West Face R2	202.5"	1.5"
18	132" 1M-1M5-1C3G1	47.5" West Face R3	47.5"	84.5"

18 other rails to purchase. Total waste: 10.25

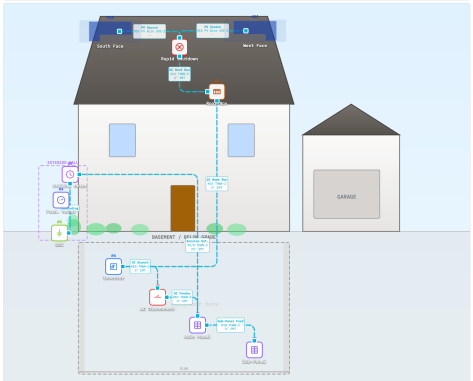
Racking Hardware BOM

ITEM	QTY	UNIT	UNIT $	LINE $
UNIRAC Rail 132" 1M-1M5-1A6-B1	8	ea	$16.25	$130.00
UNIRAC Rail 132" 1M-1M5-1C3G1	9	ea	$24.15	$216.35
UNIRAC Rail 204" 1M-1M5-2H4G1	1	ea	$41.25	$41.25
UNIRAC Mono Splice	8	ea	$4.00	$32.00
FlashFoot2	56	ea	$12.45	$697.20
UFO Mid Clamp	35	ea	$3.15	$80.50
UFO End Clamp	24	ea	$2.25	$[illegible]
UFO End Stopper 35mm	26	ea	$2.10	$12.32
WEEB General Clip	13	ea	$1.25	$12.75
Mill Grounding Lug	10	ea	$4.50	$40.20
UNIRAC End Cap Pair	20	pr	$4.15	$83.00
Double-Wire Clip	43	ea	$0.10	$4.34
			Total:	$2107.21

Site Electrical Layout

Mono Module Lunate · Residential Roof Mount · 2DW Service

+ Select | + Place | Conduit Run | + Site Run



WIRE SCHEDULE

Roof Run — Roof Run	[DC PV]	2x#10 PV Wire DWO-2
Roof Run — Inverter	[DC Home Run]	2x#14 THWN-2 + #12 Cu EGC — 1" EMT (270/480 max)
Inverter — AC Disc.	[AC Branch]	2x#10 THWN-2 + #10 THWN-2 N + #12 Cu EGC — 1" EMT (240/415 max)
AC Disc — Main Panel	[AC Feeder]	2x#10 THWN-2 + #10 THWN-2 N + #12 Cu EGC — 1" EMT (240/415 max)
Main Panel — Meter	[Service Int.]	2x#4/0 THWN-2 + #1/0 THWN-2 N + #4 Cu EGC — 2" EMT (240/415 max)
Meter — ARS	[Grounding]	#6 Bare Cu

Drag markers to move. Drag-in handles to change which edge a run connects to. 9 markers(s), 9 run(s)

EQUIPMENT

E1	Rapid Shutdown	Rapid Shutdo	NEC 690.12	E2	Roof Run	Roof Run	E3	Roof Run	Solaboard RPP90	E4	Utility Meter	Meter	Revenue
E5	Final Meter	Final Meter	Details...	E6	ARS	Grounding	Ground Rod	E7	Inverter	Inverter	Fronius Primo GEN24 4
E8	AC Disconnect	Disconnect	Lockable	E9	Main Panel	Main Panel (Existing	200A Service	E10	Sub-Panel	Panel/Breake	Basement
E11	South Face	PV Array	Details...	E12	West Face	PV Array	Details...				

CONDUIT RUNS

Run	From	To	Type	Qty	Conduit
Run 1-E1 Rapid Shutdown → E2 Roof Run	E1 Rapid Shutdo	E2 Roof Run	Type: DC	Qty: 1	Conduit: 1" EMT
Run 2-E2 Roof Run → E3 Inverter	E2 Roof Run	E3 Inverter	Type: Type 2	Qty: 1	Conduit: 1" EMT
Run 3-E3 Inverter → E4 AC Disconnect	E3 Inverter	E4 AC Disconnect	Type: AC	Qty: 10	Conduit: Type 2
Run 4-E4 AC Disconnect → E6 Main Panel	E4 AC Disconnect	E6 Main Panel	Type: Type 2	Qty: 1	Conduit: 1" EMT
Run 5-E6 Main Panel → E7 Utility Meter	E6 Main Panel	E7 Utility Meter	Type: Service Ent	Qty: 1	Conduit: 2" RMC
Run 6-E7 Utility Meter → E8 ARS	E7 Utility Meter	E8 ARS	Type: Grounding	Qty: 1	Conduit: Bare/Bare
Run 7-E8 Main Panel → E9 Sub-Panel	E8 Main Panel	E9 Sub-Panel	Type: Type 2	Qty: 1	Conduit: 1" EMT
Run 8-E12 South Face → E2 Rapid Shutdown	E12 South Face	E1 Rapid Shutdown	Type: DC	Qty: 10	Conduit: Type 2
Run 9-E11 West Face → E1 Rapid Shutdown	E11 West Face	E1 Rapid Shutdown	Type: DC	Qty: 10	Conduit: Type 2

Wire Sizing Calculations NEC 2023 Table

Segment	Wire Spec	Ampacity	NEC Check
PV Source	2x#10 PV Wire DWO-2 (DC/Roof)	Wire Ind.	• 2 conductors: OK +/- per string • Ampacity: 40A x 1.25 = 50A (NEC 690.8(A)(1)) per NEC 690.31(C) • Wp overall compliance: ~7% wiring overhead for standard runs (NEC 690.31)
DC Home Run	2x#14 THWN-2 (DC/Roof) #14 Cu EGC (DC/Roof)	1" EMT 270 / 240 max	• 2 current-carrying conductors: 40 x 1.25 (TMAX) = 50 x 0.941 (Temp Adjust) = 47.0A (Ampacity Adj) (NEC 690.8 - 240% of Isc) • Ampacity: 47.0A (Ampacity Adj) (NEC 690.8) • 2 conductors total = NEC Ch 9 Table 1: 40% Fill (1.5mm) • Conduit: 1" EMT (Type 2) (NEC 358.20) (Type 2) • Conduit: 1" EMT (358.actual) / 40% Fill (NEC Ch 9 Table 4
AC Branch	2x#10 THWN-2 (DC/Roof) #10 Cu EGC (DC/Roof) #12 Cu EGC (Conn)	1" EMT 240 / 240 max	• 2 CCC (240V split phase) + 1 ccc(s) + 1 EGC = 4 conductors • Ampacity: 40A x 1.25 = 50A (NEC 690.8(A)(1)) per output circuit • EGC: #12 AWG per NEC 250.122 for 20A OCPD • NEC Ch 9 Table 1: 40% Fill (1mm) (4 conductors) • Conduit: 1" EMT (Type 2) (NEC 358.20) (Type 2) (Existing 1) • Conduit: 1" EMT (358.actual) / 40% Fill (NEC Ch 9 Table 4
Service Entrance	2x#4/0 THWN-2 (Int/Ext) #1/0 Cu EGC (Int/Ext) #6 Cu GEC (Ext/Int)	2/0" EMT 195 / 240 max	• 200A service = #4/0 AWG Cu per NEC 310.16 / 2% Voltage Drop • Ampacity: 195A (Ampacity Adj) (NEC 310.16) (Ambient Temp) • EGC: #1/0 AWG per NEC 250.122 for 200A OCPD • Wire size: 2#4/0AWG + 1#4/0AWG = 0.5949 mm sq in • Conduit: 2/0" EMT (Type 2) (NEC 358.20) (Type 2) (Existing 2) • Conduit: 2/0" EMT (358.actual) / 40% Fill (NEC Ch 9 Table 4
GEC	#6 Bare Cu (J)	Wire Ind.	• #6 Bare Cu (J) per NEC 250.64 for 200A service • Wp overall compliance: NEC 250.64(C) - #6 AWG Cu may be run exposed

Generic site elevation showing typical equipment placement. Equipment locations are approximate - actual placement depends on site conditions, AHJ requirements, and installer preference. Residential layout shows inverter on garage/exterior wall with AC disconnect near service entrance.

Photo Studio

12 photo(s)

+ Upload | Photo Capture | & Basement Import

Exterior Wall

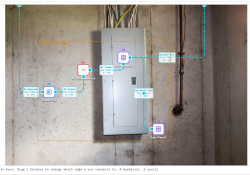
Basement

DESCRIPTION: South roof from ground...

ZONE / LOC: Basement

SOURCE: iPhone

+ Select | + Place | Conduit Run | + Site Run



Drag markers to move. Drag-in handles to change which edge a run connects to. 4 marker(s), 4 run(s)

EQUIPMENT

E1	Inverter	Inverter	Fronius Primo GEN24	E2	AC Disconnect	Disconnect	Lockable
E3	Main Panel	Panel/Breake	200A Service	E4	Sub-Panel	Panel/Breake	Basement

CONDUIT RUNS

Run	From	To	Type	Qty	Conduit
Run 1-E1 Home Run → E2 Inverter	E1 Home Run	E2 Inverter	Type: DC	Qty: 24	Conduit: Type 2
Run 2-E2 Inverter → E3 AC Disconnect	E2 Inverter	E3 AC Disconnect	Type: AC Branch	Qty: 7	Conduit: Type 2
Run 3-E3 Disconnect → E4 Main Panel	E3 AC Feeder	E4 Main Panel	Type: Type 2	Qty: 10	Conduit: Type 2
Run 4-E4 Main Panel → E4 Sub-Panel	E4 Sub-Panel	E4 Main Panel	Type: Type 2	Qty: 1	Conduit: 1" EMT
Run 5-E2 Inverter → E1 Home Run	E2 Inverter	E1 Home Run	Type: Service Ent	Qty: 1	Conduit: 2" RMC
Run 6-E2 T → E1 Inverter	E2 T	E1 Inverter	Type: Type 2	Qty: 1	Conduit: 1" EMT